

# COLBYN WADMAN

[colbyn.com](http://colbyn.com)[github.com/colbyn](https://github.com/colbyn)

801-367-4487

[hello@colbyn.com](mailto:hello@colbyn.com)

## SOFTWARE

### Instrumented systems

Crawlers, pipelines, search indexes, reporting layers. Built around the actual data model: queryable; auditable; extendable.

[Automation](#) · [Search](#) · [Data Pipelines](#) · [CLI](#) · [Reporting](#)

### Workflow automation

Manual processes converted to defined pipelines: structured inputs, quality gates, clean handoffs.

[Automation](#) · [Search](#) · [Data Pipelines](#) · [CLI](#) · [Reporting](#)

## SYSTEMS

### Architecture before components

Topology, constraints, and failure modes defined first.

[Topology](#) · [Process Design](#) · [Constraints](#) · [Documentation](#) · [Operations](#)

### Process formalization

Informal rules surfaced, named, and documented.

[Topology](#) · [Process Design](#) · [Constraints](#) · [Documentation](#) · [Operations](#)

## DESIGN

### DC lighting architecture

Distribution layout, driver selection, control protocol, dimming behavior, serviceability.

[Low Voltage](#) · [DC Distribution](#) · [Typst](#) · [Specification](#) · [Diagrams](#)

### Technical publication

Specifications, diagrams, and installation logic precise enough to build from. (Clear enough to hand to a non-specialist without a walkthrough.)

[Low Voltage](#) · [DC Distribution](#) · [Typst](#) · [Specification](#) · [Diagrams](#)

## RESEARCH

### Primary source analysis

Datasheets, standards, supplier catalogs, field constraints.

[Standards](#) · [Supply Chain](#) · [Specs](#) · [Field Constraints](#) · [Technical Writing](#)

### Actionable synthesis

Gap between specification and real installation conditions: identified, closed, documented.

[Standards](#) · [Supply Chain](#) · [Specs](#) · [Field Constraints](#) · [Technical Writing](#)

## IMPLEMENTATION

### Full-cycle execution

Layout, sourcing, commissioning, verification – without coordination loss from splitting phases across specialists who don't share a system model.

[Installation](#) · [Commissioning](#) · [Prototyping](#) · [Verification](#) · [Low Voltage](#)

### Constraint-driven prototyping

Early builds targeting the hard problem: thermal behavior, dimming continuity, distribution loss, control latency. Constraint solved before the system is committed.

[Installation](#) · [Commissioning](#) · [Prototyping](#) · [Verification](#) · [Low Voltage](#)

## ENGAGEMENT

### Where this applies

Architecture undecided. Requirements rough. Existing system producing inconsistent results with no clear cause. That's the entry point.

[Systems Thinking](#) · [Ambiguity](#) · [Technical Depth](#) · [Documentation](#) · [Handoffs](#)

### What it produces

A system with a defensible logic – documented, sourceable, buildable, and maintainable by someone other than the original designer.

[Systems Thinking](#) · [Ambiguity](#) · [Technical Depth](#) · [Documentation](#) · [Handoffs](#)

## Features

### Milestone

**2019**

Top-percentile GitHub contributor

[Source](#) <sup>1</sup>

### Highlight

#### Imager

Rust / Node image optimization algorithm using brute force heuristics

[Source](#) <sup>3</sup>

|                  |                                                                                   |                   |
|------------------|-----------------------------------------------------------------------------------|-------------------|
| source :         |  | 39.00M (4 images) |
| kraken.io :      |  | 24M               |
| jpegmini.com :   |  | 16M               |
| compression.ai : |  | 8.90M             |
| imager :         |  | 4.20M             |

### Achievement

**728+** ★

GitHub stars on most popular authored OSS tool

[Source](#) <sup>2</sup>

### Review

#### Academic

"You are such a profound writer and thinker. It has been my privilege to be your instructor of record. One day I will say, I had him in my English class. I have such high hopes for you! Go conquer your world! You're awesome."

– Dr. Jim Birrell  
[Research Project](#) <sup>4</sup>

## Early Career History

### pre 2015

#### Space Monkey

Brief High School Internship

Worked under Tom Metge

### 2016

#### Uplynk / Verizon Digital Media Services

Worked as a Jr. Developer on the QA team.

Team lead: Asiel Brumfield.

Left to work on encrypted images.

### 2015

#### Galileo Processing

Galileo Processing

Worked under Robert Raver.

### 2020

#### 2020 – UVU CS Grader

Utah Valley University

CS Grader – Computer Science

Bianca Ruiz

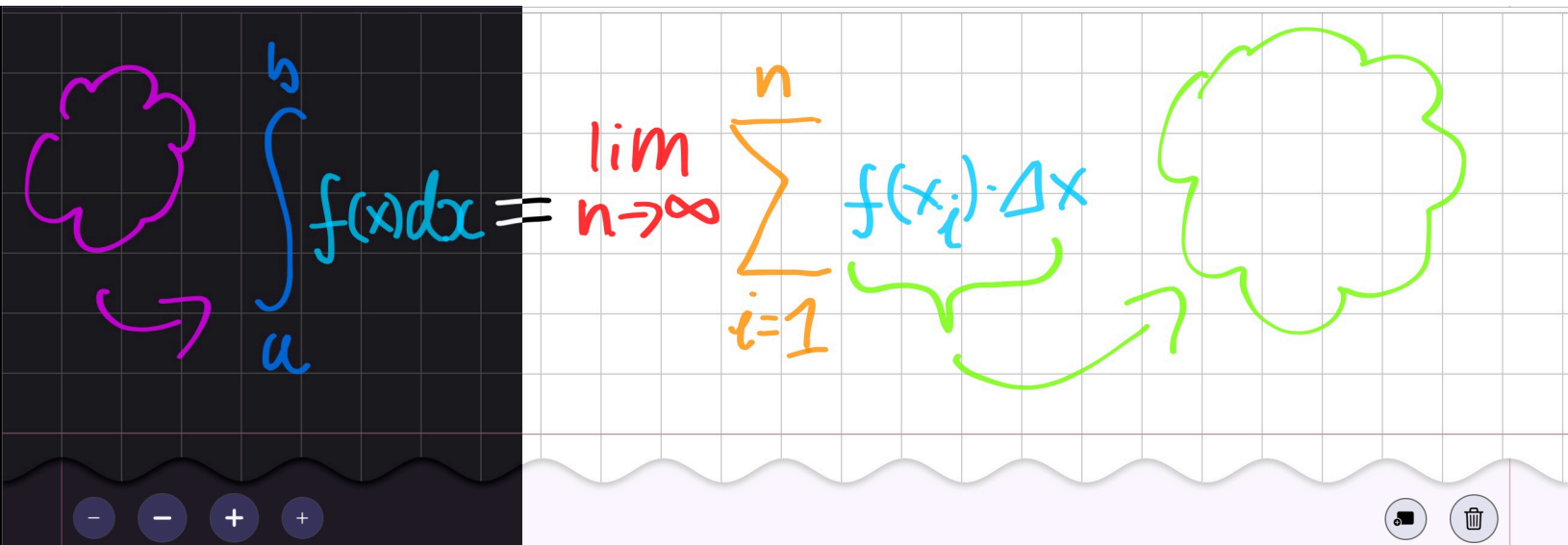
## Overview links

1 <https://github.com/colbyn?tab=overview&from=2019-12-01&to=2019-12-31>

2 <https://github.com/imager-io/imager>

3 <https://github.com/imager-io/imager>

4 <https://github.com/colbyn/analog-computing-and-refs>



## Keywords

### Infrastructure

Data center operations · mainframe buildout · rack and stack · hardware provisioning · systems support · network documentation · structured cabling · Galileo processing · ticketing · incident response · physical layer · server configuration · storage · uptime · ops

### Media & Streaming

Video streaming · multimedia · codec pipelines · encoding · transcoding · multi-bitrate · adaptive streaming · HLS · CDN · delivery infrastructure · UpLynk · Verizon Digital Media Services · latency · throughput · media processing · ingest · playback · format conversion

### Networks & Field

ISP founding · wireless infrastructure · WISP operations · last-mile · point-to-point · RF planning · subscriber management · routing · switching · TCP/IP · field installation · tower work · contracting · independent practice · sales · client acquisition · troubleshooting

### Low Voltage & Lighting

Class 2 wiring · NEC compliance · DC distribution · LED systems · driver architecture · DALI-2 · dimming · CCR · PWM · constant current · PoE · Zhaga · COB · photometry · circadian lighting · spectral design · low-voltage networking · system architecture · home run topology · power budgeting · field installation · commissioning

### Professional & Leadership

Cross-functional collaboration · stakeholder communication · project management · requirements gathering · deliverable ownership · process improvement · change management · knowledge transfer · vendor management · client-facing communication · scope definition · resource planning · risk assessment · quality assurance · agile · roadmap alignment · strategic planning · budget awareness · compliance · audit readiness

### Soft Skills & Competencies

Communication · active listening · relationship building · conflict resolution · negotiation · presentation skills · executive presence · self-directed · initiative · ownership mentality · accountability · adaptability · resilience · growth mindset · async communication · deadline-driven · detail-oriented · results-oriented · critical thinking · decision-making · prioritization · time management · problem-solving

### History

2014 : Summer Internship · Space Monkey · Decentralized P2P Cloud Computing · Infrastructure Provisioning · Systems Support

2015 : CompTIA Operations Specialist · Galileo Processing · Data Center Buildout · Hardware Provisioning · Network Documentation · Incident Response · Uptime & Ops

2015-2016 : Founder & Lead Systems Engineer · Independent WISP · Wireless Infrastructure · Last-Mile Deployment · RF Planning · Field Installation · Client Acquisition · Routing & Switching

2016 : QA Engineer, Media Systems · UpLynk / Verizon Digital Media Services · Video Streaming Pipelines · Encoding & Transcoding · HLS · CDN Delivery · Python Tooling · Throughput Optimization

2017 : Prototyping & Tooling Engineer · Eimg Startup · Video Processing Tooling · Rapid Prototyping · Multimedia Workflow Automation

2018 : Independent Software Engineer & Language Designer · Compiler Design · DSL Implementation · Rust · Haskell · Open Source Tooling

2019 : Independent Imaging Systems Developer · Imager Project · Image Processing Pipelines · FFmpeg Integration · VMAF Quality Metrics · Rust · Node.js · Performance Optimization

2020 : CS Student & Research Assistant · UVU · Academic Research · CS Grader · Subscript Publishing Toolchain · Swift · iPad Application Development

2021 : Independent DevOps & Cloud Consultant · AMI Automation · AWS Infrastructure · CI/CD Pipelines · Rust Systems Programming

2022 : Independent Researcher & Systems Developer · University of Utah · Philosophy, Writing & Rhetoric · Parser Development · Publishing Systems · Rust · Swift · Technical Documentation

2023 : Software Engineer, Publishing Systems · Subscript Publishing · iOS & macOS Development · Digital Ink & Vector Graphics · Custom Language Design · Cross-Platform Architecture

2024 : Independent Systems Engineer · Parser Combinators & AST Tooling · Dataset Generation Pipelines · Rust · Swift · Data Architecture

2025-present : Principal Consultant, Systems & Integration · Static Web Compiler · Markdown Rendering · LED & Low-Voltage Systems · DALI-2 · PoE · Technical Writing · Architecture & Client Consulting

# Spectrum Is the Real Light Source

THE HIDDEN LAYER

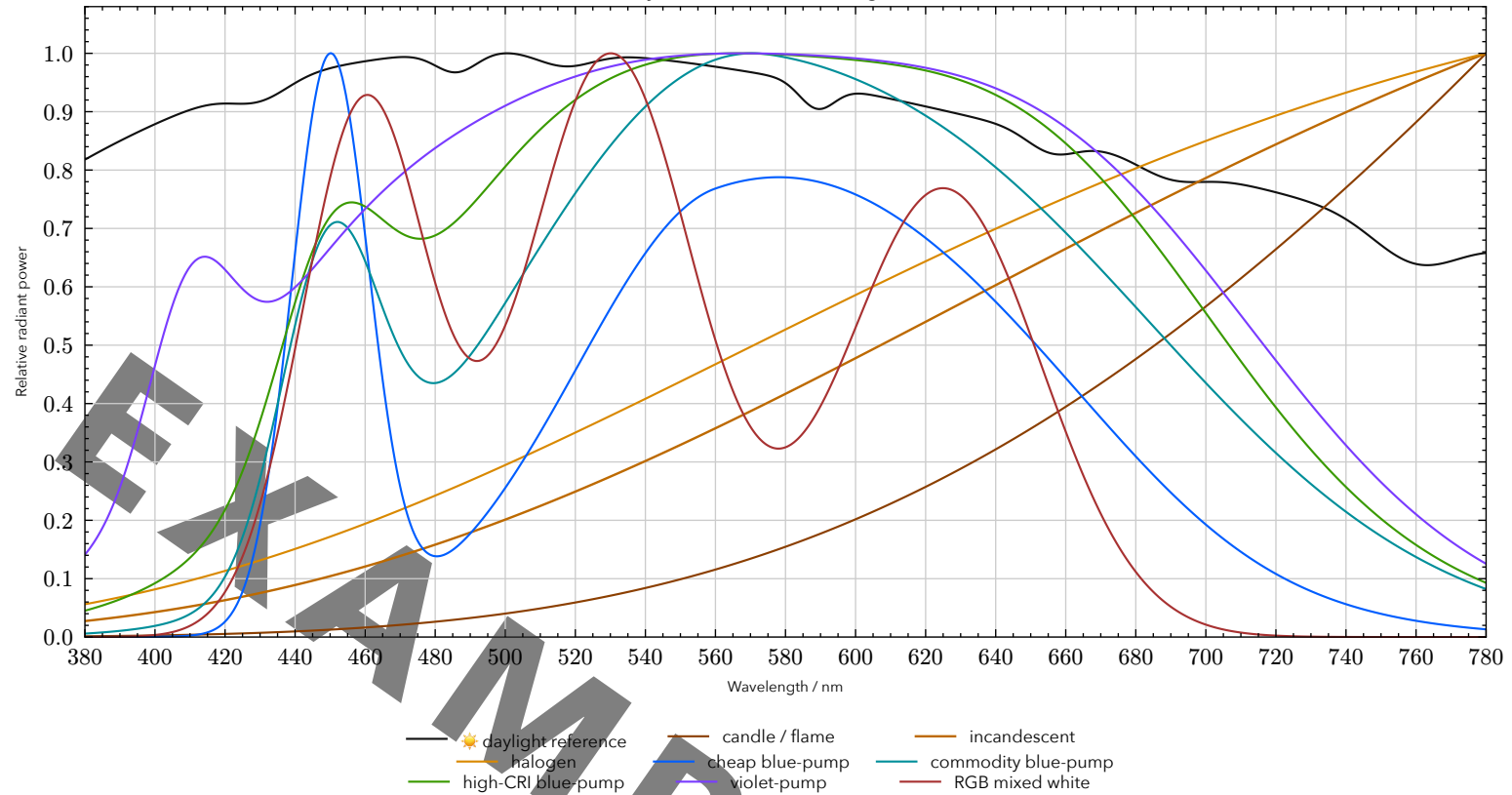
## White light is not a substance.

It is a spectral construction that happens to land on a white appearance.

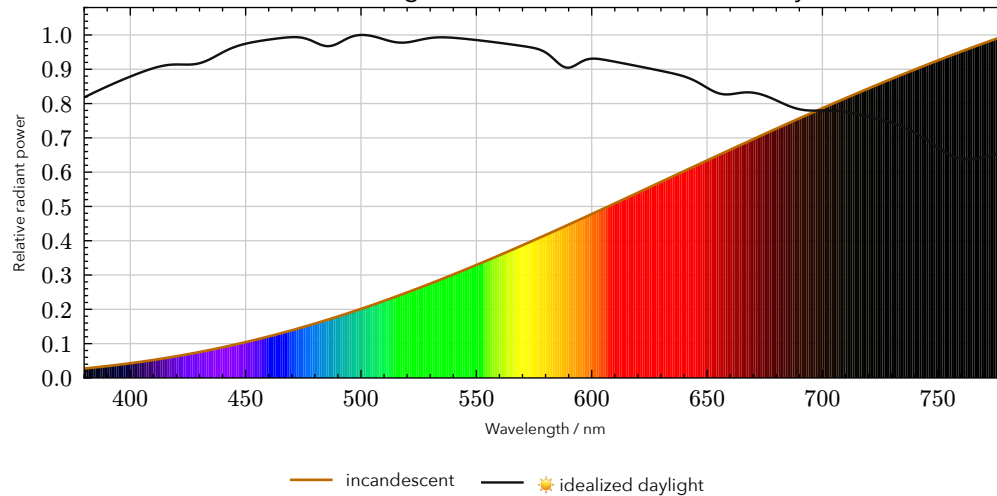
### READING RULE

Do not ask only whether the light is warm or cool. Ask where the energy is, where it is missing, and how the spectrum is made.

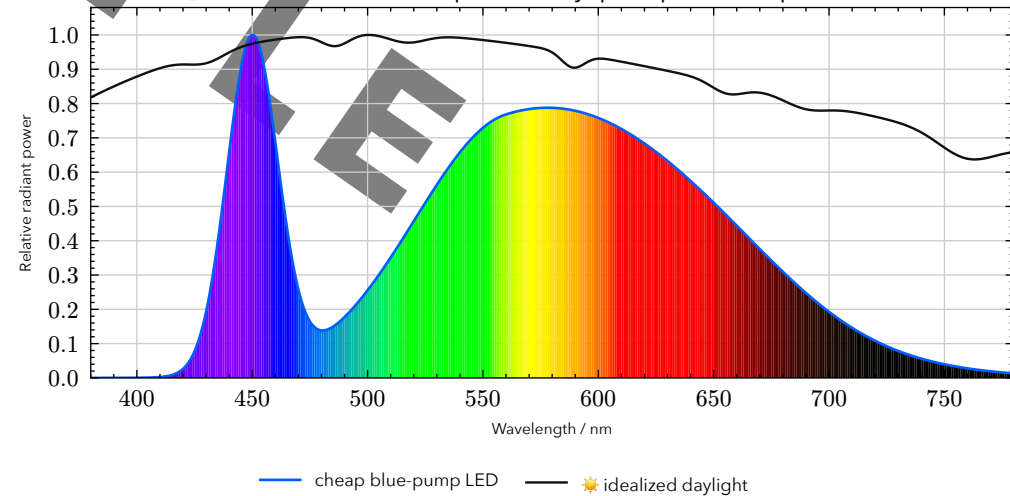
Different ways to make something that looks white



Thermal light: continuous, warm, red-heavy



Basic LED: spike, valley, phosphor hump



Incandescent is spectrally continuous, but tilted heavily toward long wavelengths. Its weakness is efficiency, not spectral assembly.

A basic blue-pump LED can look white while still having a hard pump spike, a cyan depression, and weak deep red.